



# **UPES ACM & ACM-W 100 Days of Code**

**Day: 3**

**UPES ACM & ACM-W 100 Days of Code Platform**

# Problem 1

## Plus One (LeetCode #66)

<https://leetcode.com/problems/plus-one/description/>

### Description

A problem that teaches array manipulation and carry handling by simulating addition on a large integer represented as an array.

Given an integer array `digits` where each element represents a digit of a number, increment the number by one and return the resulting array.

This problem is commonly asked in interviews and helps build concepts like:

- carry propagation
- array traversal
- edge case handling
- number simulation

which are important for mathematical and array-based problem solving.

**Your Task:** Add one to the number represented by the array using an optimized approach.

### Input

An integer array `digits` where each element is a single digit (0–9).

Constraints:  $1 \leq n \leq 100$ ,  $0 \leq \text{digits}[i] \leq 9$

### Output

Print the resulting array after incrementing the number by one.

| Sample Input 1                | Sample Output 1      |
|-------------------------------|----------------------|
| <code>digits = [1,2,3]</code> | <code>[1,2,4]</code> |

### Explanation

The array `[1, 2, 3]` represents the integer **123**. Adding one gives **124**, returned as `[1, 2, 4]`.

For the edge case `[9]`, the number is **9**. Adding one gives **10**, which requires a new leading digit, returned as `[1, 0]`.

| Sample Input 2                  | Sample Output 2        |
|---------------------------------|------------------------|
| <code>digits = [4,3,2,1]</code> | <code>[4,3,2,2]</code> |

| Sample Input 3            | Sample Output 3    |
|---------------------------|--------------------|
| <code>digits = [9]</code> | <code>[1,0]</code> |

# Problem 2

## Number of Occurrences

### Description

A problem that teaches binary search and array traversal techniques by finding how many times a target element appears in a sorted array.

Given a sorted array `arr[]` and a target value, return the total number of occurrences of the target in the array.

This problem is commonly asked in interviews and helps build concepts like:

- searching in sorted arrays
- range finding
- optimized traversal

which are important for efficient searching problems.

**Your Task:** Find the number of occurrences of the target element using an optimized approach.

### Input

An integer sorted array `arr[]` and a target value.

Constraints:  $1 \leq n \leq 10^6$ ,  $-10^9 \leq arr[i], target \leq 10^9$

### Output

Print the total number of occurrences of `target` in the array.

| Sample Input 1                                                  | Sample Output 1 |
|-----------------------------------------------------------------|-----------------|
| <code>arr[] = [1,1,2,2,2,2,3]</code><br><code>target = 2</code> | 4               |

| Sample Input 2                                                  | Sample Output 2 |
|-----------------------------------------------------------------|-----------------|
| <code>arr[] = [1,1,2,2,2,2,3]</code><br><code>target = 4</code> | 0               |

| Sample Input 3                                                     | Sample Output 3 |
|--------------------------------------------------------------------|-----------------|
| <code>arr[] = [8,9,10,12,12,12]</code><br><code>target = 12</code> | 3               |

### Explanation

The element 2 appears 4 times in the array `[1, 1, 2, 2, 2, 2, 3]`, so the output is **4**.

If the target 4 is not present in the array, the output is **0**.